Magnetic fields in solar system - a review

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Introduction

A stable, sufficiently strong magnetosphere appears to be necessary for a planet to retain its atmosphere and remain habitable over (long) geological period of time. Ganymede, Jupiter' largest moon (and also the largest natural satellite in the solar system; bigger than Mercury) is one of the few moons known to have its own magnetic field. Its field is strong enough to create a distinct boundary between the Jovian field and its own. Its magnetic dipole moment (vector quantity describing overall strength & orientation) is also 3 times that of the magnetic moment of Mercury. On the other side, magnetic fields also influence star formation. In interstellar clouds, magnetic fields help in resisting total gravitational collapse (Carr, et al., 1969). So magnetic field slow the whole process. If there was no magnetism, star formation rates in the galaxy would probably be higher.

The solar magnetic field does not terminate at the surface of the Sun. It is carried outward by the solar wind plasma into interplanetary space, forming the interplanetary magnetic field (IMF). The Voyager spacecraft, now beyond the heliopause (the rough boundary where the solar wind ends) at over 150 AU, have directly measured the transition from the solar-dominated field to the local interstellar magnetic field, which is oriented at roughly 30 degrees to the galactic plane and has a strength of around $3X10^{-6}$ Gauss.

The speed and density of the solar wind, along with the orientation of the IMF affect literal space throughout the solar system. Coronagraphs are instruments which can be used to remotely track Coronal Mass Ejections (CMEs). And occasional radio emissions reveal indirect signatures on particles and ion streams which could be linked back to magnetic field flows.

Understanding charged particle motion

Electrons and ions spiral around magnetic field lines, in helical paths. It is an intrinsic emergent property due to the field itself, and applies to all particles interacting with it. If the field is strong enough, it can trap high-energy particles forming belt-like regions like in the case of Earth and Jupiter.

A planetary magnetic field can act as a protective bubble against solar wind and cosmic rays. Therefore a strong and stable magnetosphere could be a necessary component for life-bearing planets. Mars used to have a weak magnetic field but it has been stripped away now. Though Earth's field is moderate in strength, it is adequate to shield it from the strong solar wind and cosmic rays that could strip away its atmosphere. On the other hand, Jupiter's field is so strong that it influences the orbital environment of its moons.

Charged particles experience a force

$$\mathbf{F} = q(\mathbf{v} \times \mathbf{B})$$

that makes them spiral around field lines and move perpendicularly relative to the field. **B** is the magnetic field and **q** is the charge. **v** is velocity, which is perpendicular to both the field lines and the charged particle.

Synthesis

The dominant magnetic structure affecting the solar system is from the Sun. The field is dragged outward due to the solar wind (~120 AU). But the solar system itself is moving through something called the Local Interstellar Cloud, which has its own magnetic field. The cloud is composed of gas, dust and plasma and is about 30 light years across.

The interaction between the solar wind and this material from the outside, so to speak, creates a boundary called the heliopause. The heliopause is where the domination of

the solar wind ends. Voyager 1 and Voyager 2 crossed this boundary (in 2012 & 2018, respectively). Both of the spacecraft observed that the magnetic field did not rotate *majorly*. It was expected that the direction of the solar-dominating field would change to orient itself with that of the interstellar cloud (M. Opher et al., 2013). But instruments on both the spacecraft (which observe plasma & cosmic ray particles) measured almost no rotation as they crossed the heliopause (L. Burlaga et al., 2013). The field kept pointing in the same direction as the solar field.

There are simply many competing explanations among different camps. M. Opher et al. (2013, 2017) and D. Turner et al. (2024) explain that due to processes like magnetic reconnection, the predicted change in the angles is actually very small, the same as that was observed. L. Fisk et al. (2014) cast doubt on the spacecraft themselves, doubting whether Voyager 1 actually crossed the heliopause. And finally J. Grygorczuk et al. (2014) propose that both the fields pointing in the same direction is simply a coincidence.

Conclusion

Active magnetic shielding, rarely used for the existence of other more reliable methods, uses superconducting coils to generate fields capable of deflecting charged particles from solar storms and a fraction of galactic cosmic rays before they ever reach the insides of the craft. But this method is not enough, which is why it is rarely used (Elias, et al., 2022).

Materials rich in hydrogen like polyethylene are a more effective countermeasure. So from a practical point of view: magnetic systems to reduce the incoming flux, hydrogen-dense structural materials to absorb what remains, and spacecraft layouts that place instruments and computers strategically to further lower dosage. This is also the way to go for manned spacecraft missions in the future, although for humans safety the constraints have to be even stronger.

Mission planners and engineers use

- Passive shielding
    - allocate shielding to sensitive electronics & detectors
- Specific architectural design
    - minimizes risk to essential instruments & computers
- Orbit
    - highly elliptical
    - schedule fewer passes through radiation belts

Single-point failures are identified and are either removed or given extra protection. Replacement designs are hardened for radiation.

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